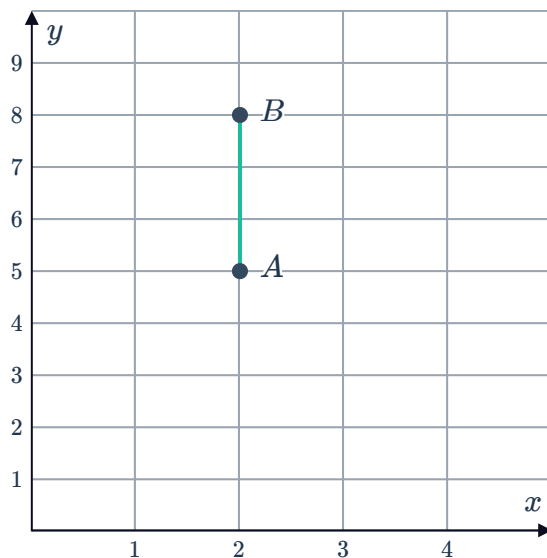


Distance between two points

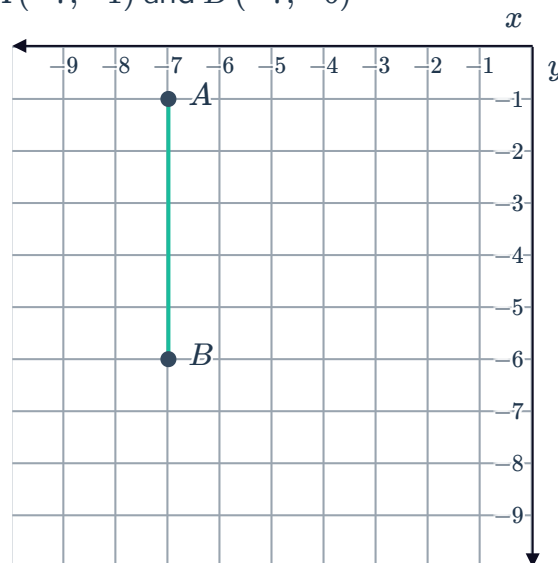


1 Find the length of each vertical interval on the number planes:

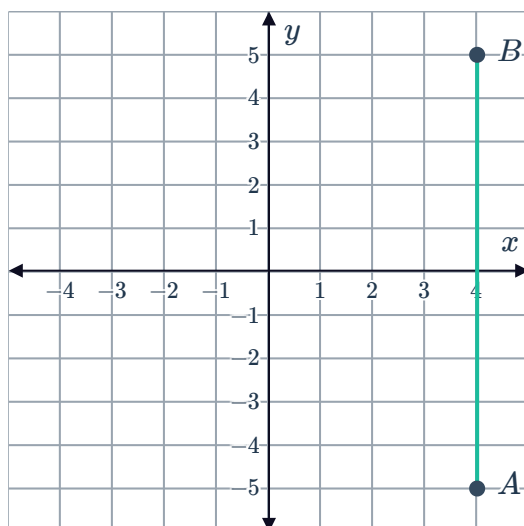
a $A(2, 5)$ and $B(2, 8)$



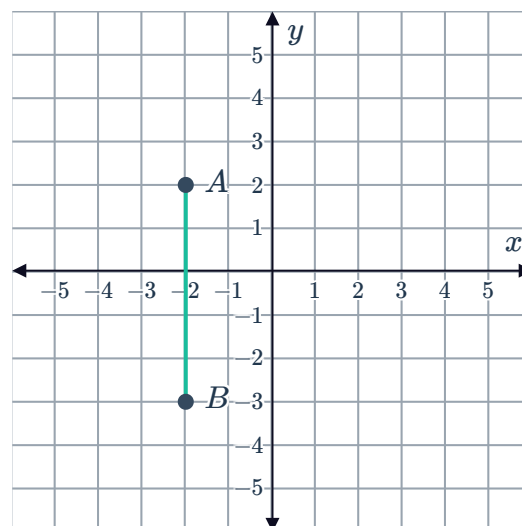
b $A(-7, -1)$ and $B(-7, -6)$



c $A(4, -5)$ and $B(4, 5)$



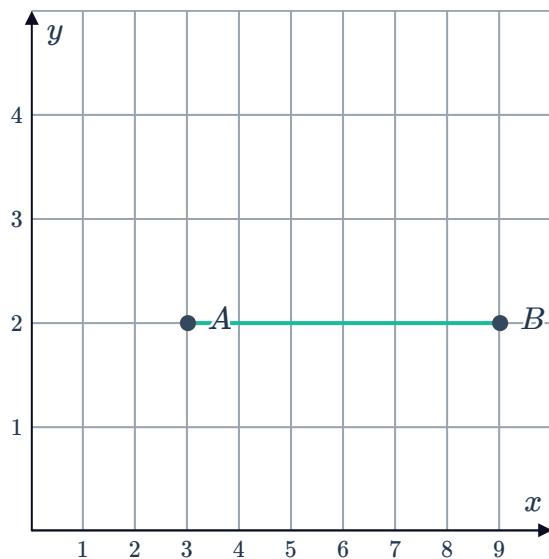
d $A(-2, 2)$ and $B(-2, -3)$



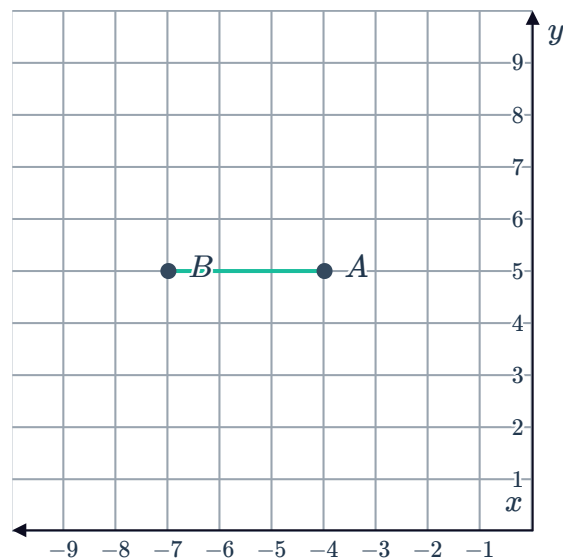
2 Find the length of each horizontal interval on the number planes:



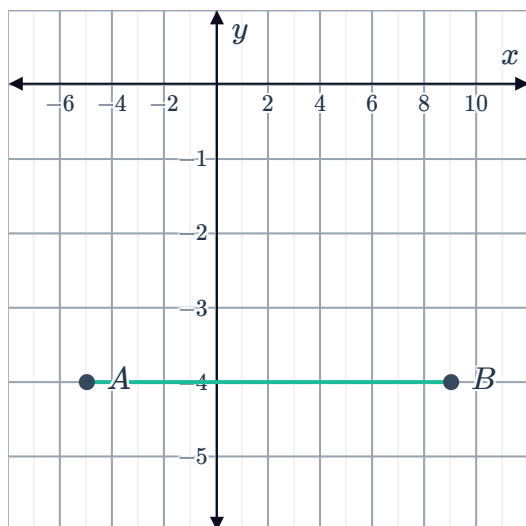
a $A(3, 2)$ and $B(9, 2)$



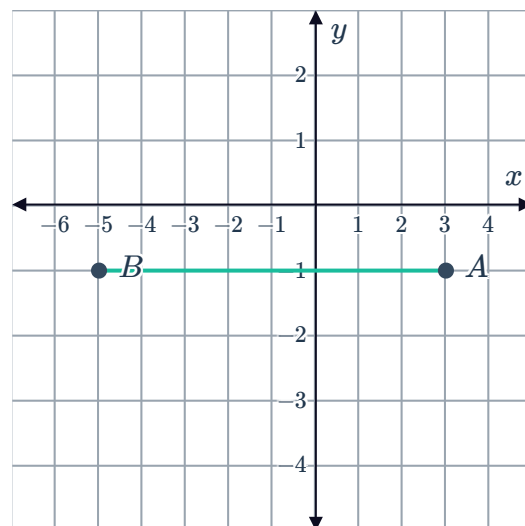
b $A(-4, 5)$ and $B(-7, 5)$



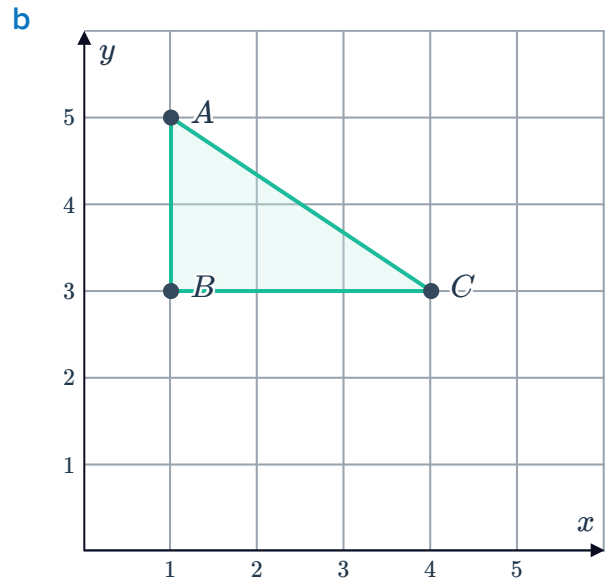
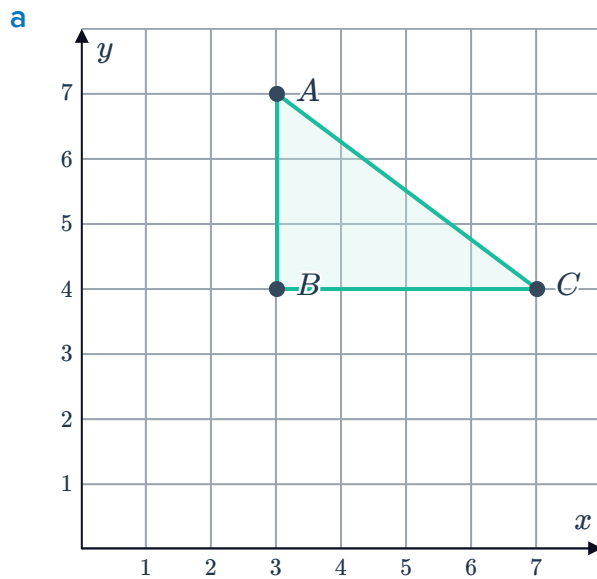
c $A(-5, -4)$ and $B(9, -4)$



d $A(3, -1)$ and $B(-5, -1)$



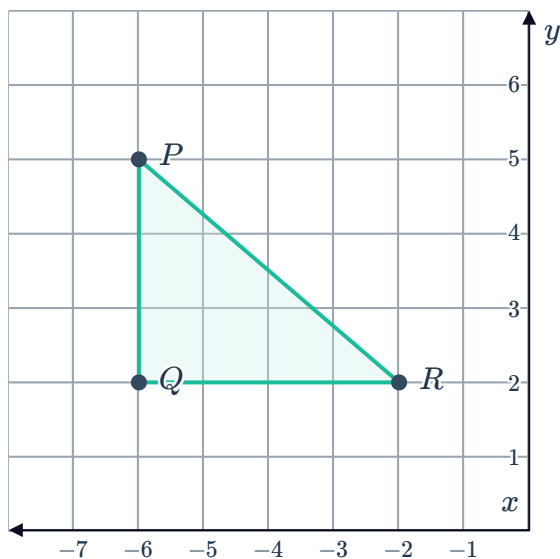
- 3 For each of the following right-angled triangles  find the length of side AC . Round your answer to two decimal places if necessary.



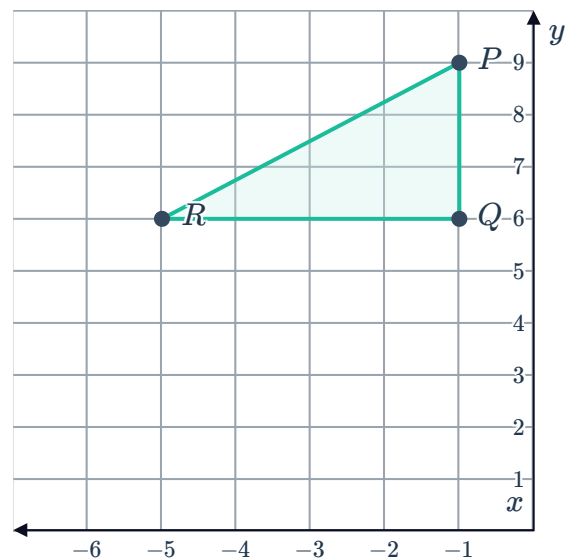
- 4 For each of the following right-angled triangles:

- Find the length of interval PQ .
- Find the length of interval QR .
- Find the length of PR , rounding to two decimal places if necessary.

- a $P(-6, 5)$, $Q(-6, 2)$ and $R(-2, 2)$



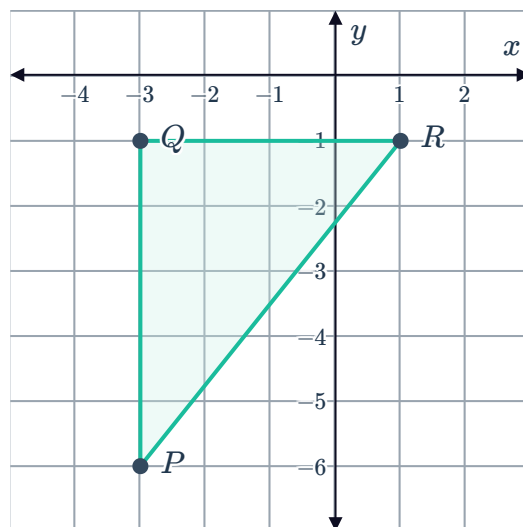
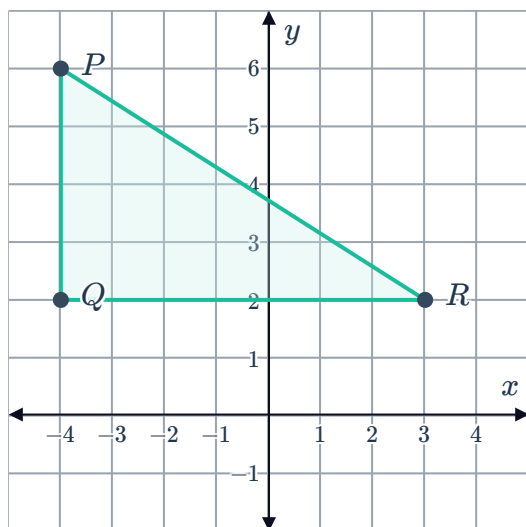
- b $P(-1, 9)$, $Q(-1, 6)$ and $R(-5, 6)$



c $P(-4, 6)$, $Q(-4, 2)$ and $R(3, 2)$



d $P(-3, -6)$, $Q(-3, -1)$ and $R(1, -1)$

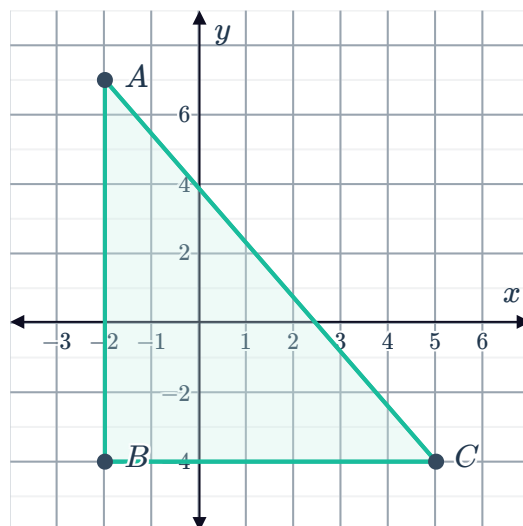
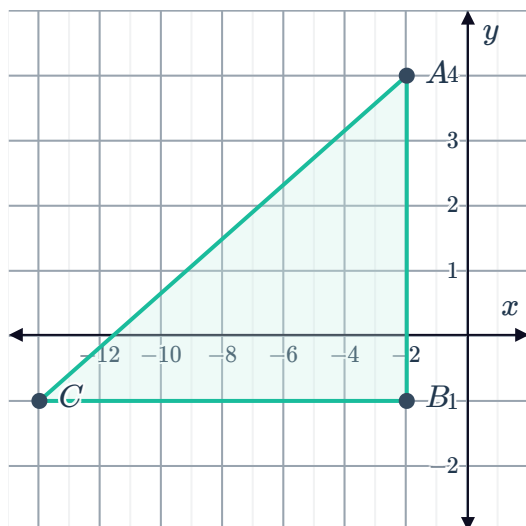


5 For each of the following right-angled triangles:

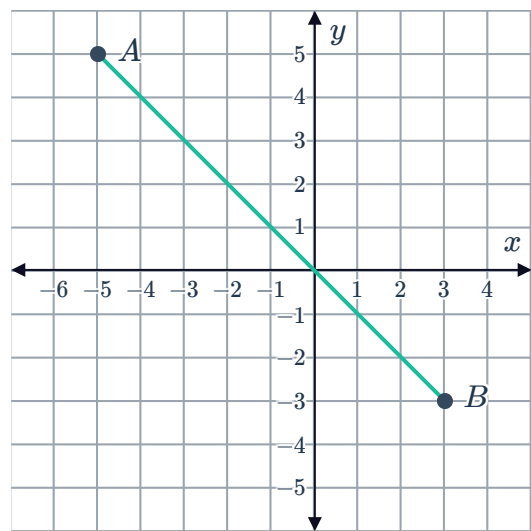
- i Find the length of interval AB .
- ii Find the length of interval BC .
- iii Find the length of AC denoted by c , rounding to two decimal places if necessary.

a $A(-2, 4)$, $B(-2, -1)$ and $C(-14, -1)$

b $A(-2, 7)$, $B(-2, -4)$ and $C(5, -4)$



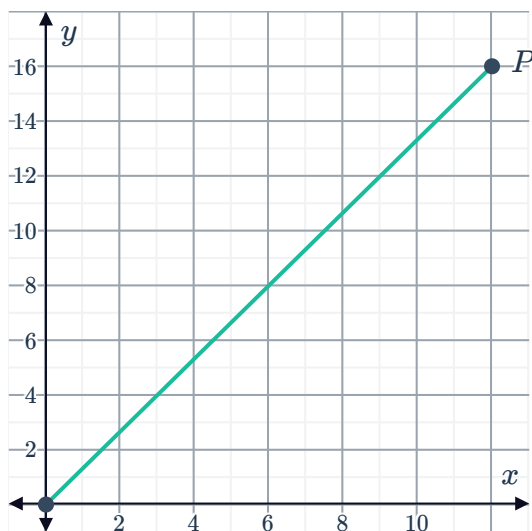
- 6 Find the length of AB shown on the graph. Round your answer to two decimal places.



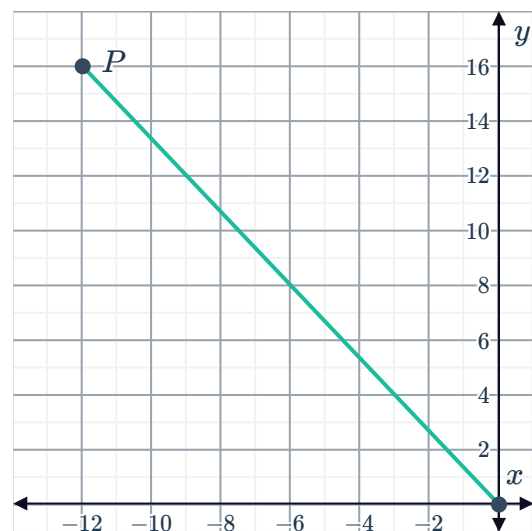
Distance formula (Extended)

- 7 Find the distance of the given point P from the origin.

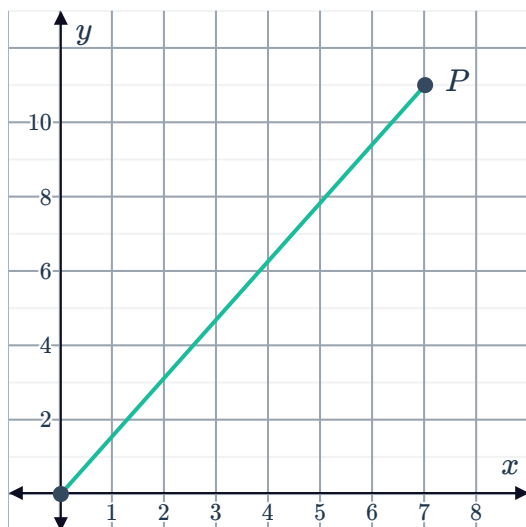
a $P(12, 16)$



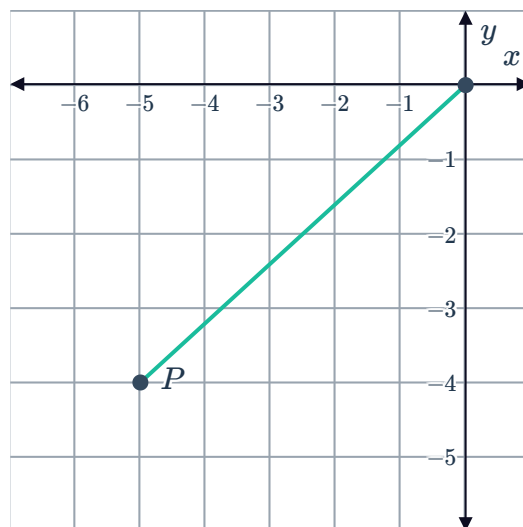
b $P(-12, 16)$



c $P(7, 11)$

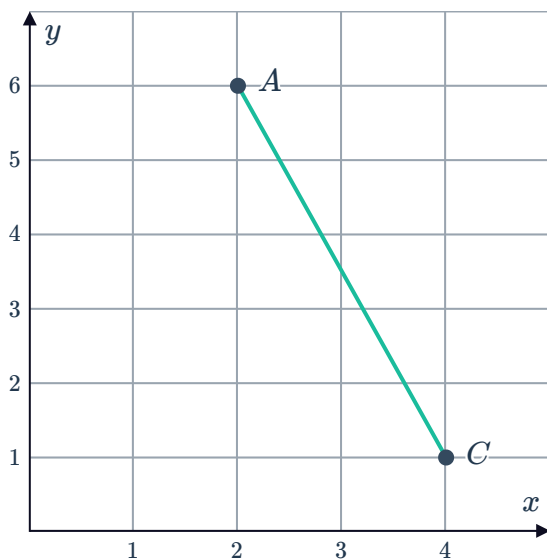


d $P(-5, -4)$

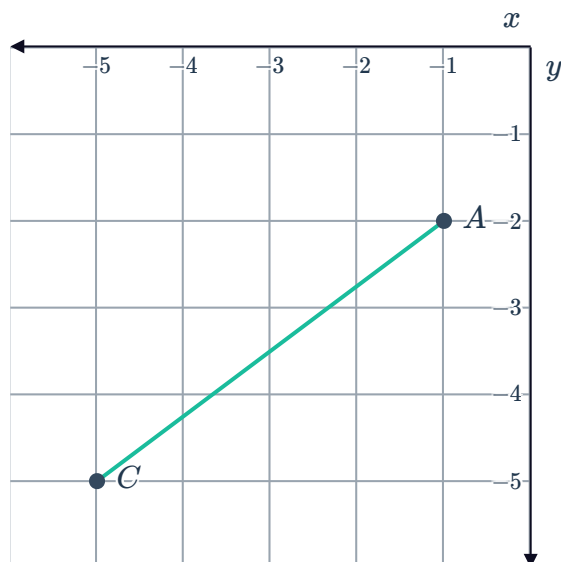


8 For each of the following graphs, find the length of the interval AC . Round your answer to two decimal places if necessary:

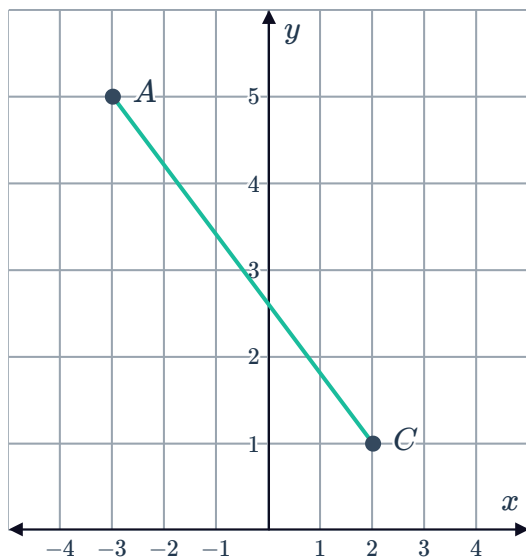
a



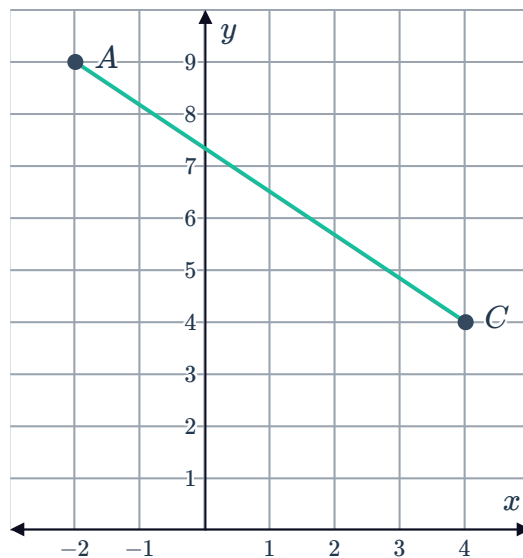
b



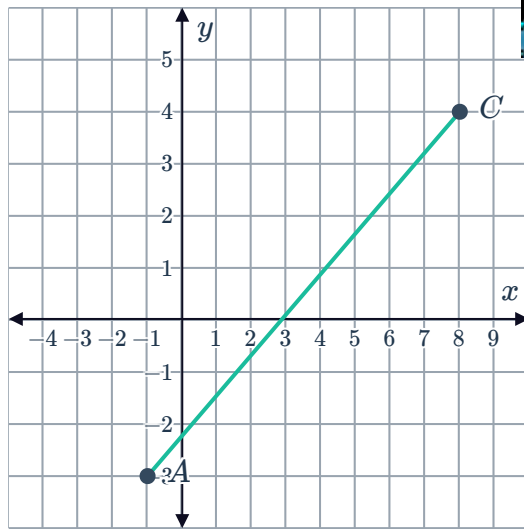
c



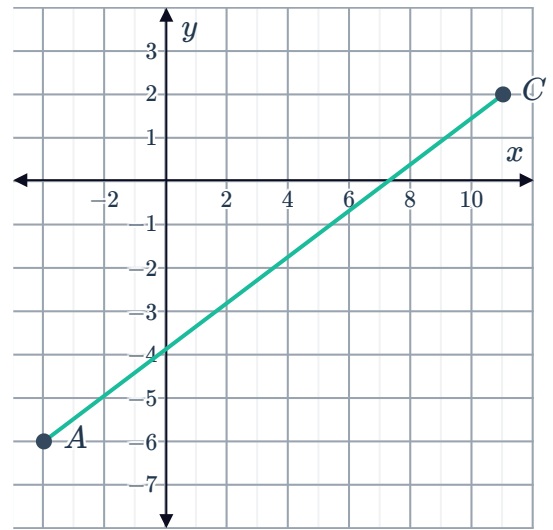
d



e



f



9 Find the distance between Point A and Point B . Write your answer in surd form if necessary.

a $A(1, 4)$ and $B(7, 12)$

b $A(4, 2)$ and $B(-8, -7)$

c $A(-1, 9)$ and $B(-4, 1)$

d $A\left(-1, -\frac{3}{5}\right)$ and $B\left(4, \frac{12}{5}\right)$

10 Consider the points $M(-9, -1)$ and $N(1, 5)$.

a Find the exact distance from M to the origin.

b Find the exact distance from N to the origin.

c Which point is closer to the origin?

11 Given $P(4, 3)$, $M(-3, -4)$ and $N(-7, 1)$.

a Find the distance from P to M to two decimal places.

b Find the distance from P to N to two decimal places.

c Which point is further from P ?

12 Consider the Points $A(12, 3)$ and $B(14, 0)$.

a Find the length of AB to two decimal places.

b If M is the midpoint of AB , find the length of AM to two decimal places.

13 AM is a vertical interval 3 units long. If A is the point $(-2, 6)$, find two possible coordinates of M .

14 Consider the points $A(-2, -6)$, $B(4, -2)$ and $C(1, -4)$. Find:



- a The exact distance AC .
- b The exact distance BC .
- c What do you notice about the distance between the points?

15 K is the midpoint of $A(3, 1)$ and $C(15, -7)$. Find the distance from A to K correct to one decimal place.

Geometric applications (Extended)

16 A triangle is formed by three points: $A(4, -3)$, $B(1, 0)$ and $C(7, 0)$. Find the following:

- a The distance BC .
- b The exact distance AB .
- c The exact distance AC .
- d Is this triangle equilateral, isosceles or scalene?

17 $ABCD$ is a rhombus whose vertices are $A(1, 2)$, $B(3, 10)$, $C(11, 12)$ and $D(9, 4)$. Find:

- a The exact length of diagonals:
 - i AC
 - ii BD
- b The exact area of the rhombus.

18 A circle with centre at point $C(3, 4)$ has point $A(-12, 12)$ lying on its circumference. Find:

- a The radius of the circle.
- b The exact circumference of the circle.
- c The distance between point $(20, 4)$ and the centre.
- d Does the circle also pass through the point $(20, 4)$? Explain your answer.

19 A triangle has vertices at $A(1, -1)$, $B(-3, -4)$ and $C(5, -4)$.

- a Find the length of the following sides:
 - i AB
 - ii AC
 - iii BC
- b Is this triangle equilateral, isosceles or scalene?

20 Consider the triangle whose vertices are  $A(-2)$, $B(2, -8)$ and $C(3, 5)$.

- a** How can we show that the triangle has a 90° and two 45° angles?
- b** Find the length of the following sides in exact form:
 - i** AB
 - ii** AC
 - iii** BC
- c** Does the triangle have a 90° and two 45° angles?

21 $\triangle PQR$ has vertices $P(2, -6)$, $Q(-9, -17)$ and $R(-5, 1)$.

- a** Find the length of the following sides in exact form:
 - i** PQ
 - ii** QR
 - iii** PR
- b** Is the triangle right-angled?

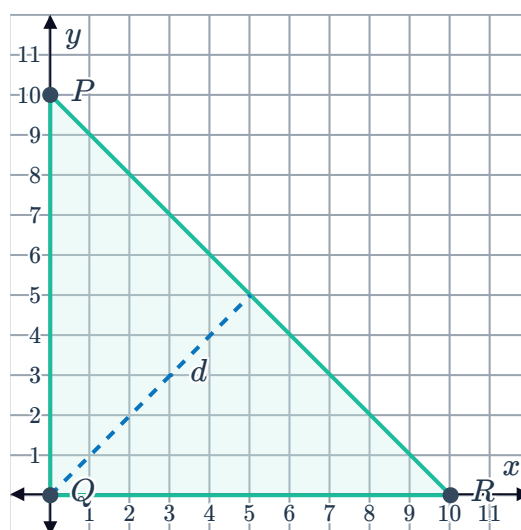
22 The points $M(-3, -5)$ and $R(4, -12)$ are the end points of an interval. $N(-1, -7)$ is a point on this interval.

- a** Find the exact distance between the following points:
 - i** M and N
 - ii** N and R
- b** In what ratio does N divide MR ?

23 Find the perimeter of the parallelogram whose vertices are $A(3, 2)$, $B(-2, 7)$, $C(2, 9)$ and $D(7, 4)$. Round your answer correct to one decimal place.

24 The isosceles $\triangle PQR$ is shown on the number plane:

- a** Find the area of the triangle.
- b** Find the exact length of PR .
- c** Find the exact value of d , the perpendicular distance from Q to the side PR .



25 Consider the points $P(10x, -6x)$ and $Q(4x, -2x)$, where $x > 0$.

- a** Find the distance between P and Q in terms of x .
- b** Find the coordinates of the midpoint of the segment PQ in terms of x .

26 $P(-4, -4)$, $Q(5, -7)$, $R(9, -11)$ and $S(0, -8)$ are the vertices of a quadrilateral.

- a** Find the exact length of side PQ .
- b** Find the exact length of side QR .
- c** Find the exact length of side RS .
- d** Find the exact length of side SP .
- e** Find the exact length of the diagonal PR .
- f** Find the exact length of the diagonal QS .
- g** Classify the quadrilateral.